

Demonstrating Magnus: A Magnetic Hand Exoskeleton for Fast and Dexterous Finger Actuation

Jun Nishida

University of Maryland, College Park
Maryland, United States
Sony Computer Science Laboratories, Inc.
Tokyo, Japan
jun@umd.edu

Yasuko Namikawa

Sony Computer Science Laboratories, Inc.
Tokyo, Japan
namikawa@golem.iit.tsukuba.ac.jp

Daisuke Tajima

Sony Computer Science Laboratories, Inc.
Tokyo, Japan
daisuke.tajima@csl.sony.co.jp

Shunichi Kasahara

Sony Computer Science Laboratories, Inc.
Tokyo, Japan
Okinawa Institute of Science and Technology
Okinawa, Japan
kasahara@csl.sony.co.jp

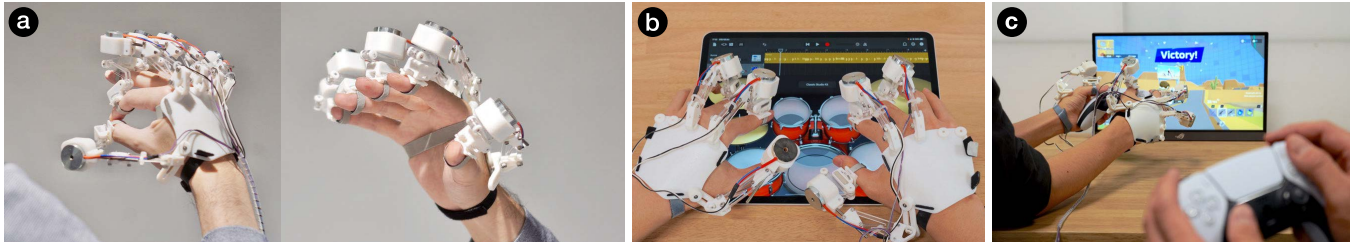


Figure 1: (a) We developed a magnetic hand exoskeleton that can provide high-speed finger actuation, allowing to assist (b) fast finger drumming and (c) eSports

ACM Reference Format:

Jun Nishida, Daisuke Tajima, Yasuko Namikawa, and Shunichi Kasahara. 2024. Demonstrating Magnus: A Magnetic Hand Exoskeleton for Fast and Dexterous Finger Actuation. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641517.3664390>

1 INTRODUCTION

Human-computer integration is an emerging interaction paradigm and has been extended to physical interactions (i.e. a computer physically actuates a user's limbs to perform a task together) by means of motor-based exoskeletons [Saraiji et al. 2018] and ones using electrical muscle stimulation (EMS) [Lopes et al. 2015; Nishida et al. 2017], accelerating motor skill learning such as that of musical instruments and sports. While these force-feedback systems demonstrated promising results in motor learning, they face challenges when applied to more timing-sensitive scenarios such as

gaming or drumming. For instance, a set of gears in motor-based systems lowers the maximum actuation speed as well as the maximum frequency of body actuation, which is required to perform fast finger-tapping motion. The gears also reduce back-drivability, preventing users from freely moving their fingers. While EMS can directly actuate muscles with small electrodes and batteries, it induces notable pain and muscle fatigue when attempting to generate high-speed body movements by applying higher stimulation voltage on the skin. Furthermore, due to the nature of the muscle-skeletal system and muscle tissues, there is a delay between the stimulation and the actual muscle or tendon contraction (measured as 56.4ms), limiting its range of applications in timing-sensitive scenarios.

1.1 Magnus: A Electromagnet-powered Hand Exoskeleton

To tackle these challenges in force-feedback systems, we designed and implemented a new form of a hand exoskeleton, called *Magnus*, which is powered by electromagnets that can provide fast, dexterous, and yet flexible finger actuation (Figure 1a). Since the Magnus exoskeleton uses pairs of electromagnets and permanent magnets, it can provide fast force-feedback (measured as 11.7ms in our evaluation). This allows for providing a higher frequency of flexing and extending force on finger joints compared to EMS. Additionally, due to the nature of magnetic actuation, it provides no pain and excessive fatigue unlike EMS, and has significantly

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SIGGRAPH Emerging Technologies '24, July 27–August 01, 2024, Denver, CO, USA
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ACM ISBN 979-8-4007-0524-3/24/07
<https://doi.org/10.1145/3641517.3664390>

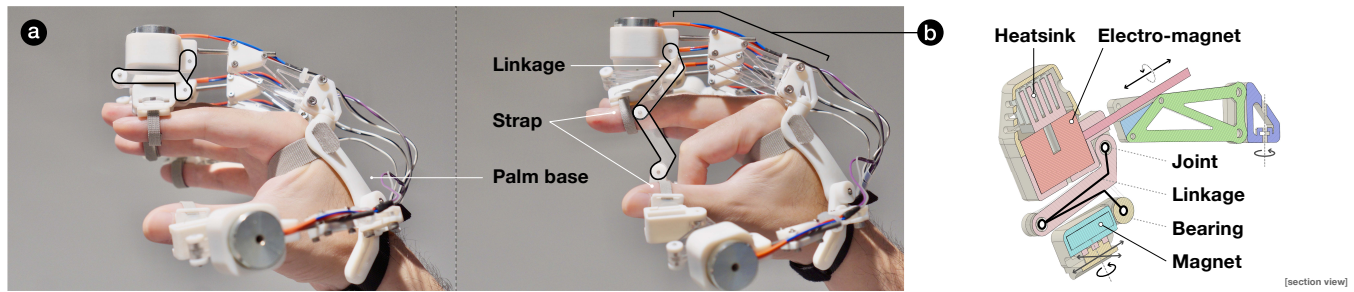


Figure 2: (a) Overview of the exoskeleton configuration (b) Section view of the magnetic actuation mechanism

higher backdrivability compared to motor-based actuation systems. When the electromagnets are not activated, they do not generate additional force unlike gears, allowing the user to move their fingers freely without feeling any resistance. We configured all these functions in a wearable form factor implemented on the back of the hand, enabling users to use it with general user interfaces, everyday tools, and even musical instruments.

1.2 Related Works

Numerous force-feedback systems have been proposed in recent years. Canonical examples include exoskeletons equipped with geared motors. These systems are capable of providing ample force to actuate the human body through linkages and gears [Saraiji et al. 2018]. Researchers also investigated new ways to use magnetic components for adding vibrotactile sensation and force-feedback [Langerak et al. 2020; McIntosh et al. 2019] to objects, thereby enhancing tangible interactions. A tabletop equipped with internal electromagnets has enabled the transfer of keyboard press motions over a distance [Tajima et al. 2022].

2 IMPLEMENTATION

In order to provide magnetic force-feedback to individual finger joints, we designed and implemented a hand exoskeleton, as shown in Figure 2 (a). This exoskeleton is comprised of 1) electromagnets, 2) mechanical linkages, 3) permanent magnets, 4) heatsinks, and 5) a palm base.

Figure 2 (b) represents the detailed structure of the actuation mechanism. We added a connection linkage between the electro-magnet and the permanent magnet to ensure a correct initial position and trajectory of the permanent magnet every time a user bends and flexes a finger, as shown in Figure 2 (a). We also wanted to ensure that the user could freely move fingers with minimum restriction, thus three adjustable joints and three free axes have been added to absorb positional and rotational displacements caused while bending a finger.

An electromagnet (uxcell 46g/0.1lb, diameter: 25mm, height: 20mm) is used for each finger actuation. The electromagnets are controlled by a microcontroller (Arduino Nano 33 BLE, Arduino) through motor driver boards (MDD10A, Cytron Technologies), which are provided with 25V from a power supply unit. The weight of the exoskeleton is 280g with three fingers. With five fingers, it weighs 452g in total.

We measured actuation property by using a force gauge (Microbeam-type load cell SC133 with an AD converter HX711), sampling at a rate of 200Hz. It was measured that it took 11.7ms until it reached peak force (0.5N), providing an impulsive force of $0.66 \times 10^{-2} \text{N}\cdot\text{s}$, while with EMS it took 56.4ms until it reached peak force (0.49N) to provide an impulsive force of $6.2 \times 10^{-2} \text{N}\cdot\text{s}$.

3 VISITOR EXPERIENCE

Our first demo scenario involves **fast finger drumming** (Figure 1b), assisted by the exoskeleton. Visitors can experience playing complex and high-speed finger drum sequences beyond their original capabilities, while minimizing the sense of intervention from the device compared to other force-feedback systems. Our second scenario extends to interpersonal high-speed gesture communication between two visitors. A pair of visitors will engage in a **first-person shooting game** that requires quick decision-making and finger actions (Figure 1c). These visitors can experience combined body abilities, such as one visitor assisting the other in shooting so that the latter can concentrate on aiming.

REFERENCES

- Thomas Langerak, Juan José Zárate, David Lindbauer, Christian Holz, and Otmar Hilliges. 2020. Omni: Volumetric Sensing and Actuation of Passive Magnetic Tools for Dynamic Haptic Feedback. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*. ACM, Virtual Event USA, 594–606. <https://doi.org/10.1145/3379337.3415589>
- Pedro Lopes, Patrik Jonell, and Patrick Baudisch. 2015. Affordance++: Allowing Objects to Communicate Dynamic Use. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. ACM, Seoul Republic of Korea, 2515–2524. <https://doi.org/10.1145/2702123.2702128>
- Jess McIntosh, Paul Strohmeier, Jarrod Knibbe, Sebastian Boring, and Kasper Hornbæk. 2019. Magnetips: Combining Fingertip Tracking and Haptic Feedback for Around-Device Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland Uk, 1–12. <https://doi.org/10.1145/3290605.3300638>
- Jun Nishida, Shunichi Kasahara, and Kenji Suzuki. 2017. Wired Muscle: Generating Faster Kinesthetic Reaction by Inter-Personally Connecting Muscles. In *ACM SIGGRAPH 2017 Emerging Technologies*. ACM, Los Angeles California, 1–2. <https://doi.org/10.1145/3084822.3084844>
- Mhd Yamen Saraiji, Tomoya Sasaki, Reo Matsumura, Kouta Minamizawa, and Masahiko Inami. 2018. Fusion: Full Body Surrogacy for Collaborative Communication. In *ACM SIGGRAPH 2018 Emerging Technologies*. ACM, Vancouver British Columbia Canada, 1–2. <https://doi.org/10.1145/3214907.3214912>
- Daisuke Tajima, Taku Tanichi, Mohammad H. Shehata, and Shunichi Kasahara. 2022. Low-Latency Motion Transfer with Electromagnetic Actuation for Joint Action. In *SIGGRAPH Asia 2022 Emerging Technologies*. ACM, Daegu Republic of Korea, 1–2. <https://doi.org/10.1145/3550471.3558399>